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Characterizing the variability of food waste quality: A need for efficient valorisation through anaerobic digestion

Henry Fisgativa, Anne Tremier *, Patrick Dabert

Irstea, UR GERE, 17 avenue de Cucillé, CS 64427, F-35044 Rennes Cedex, France

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ABSTRACT

In order to determine the variability of food waste (FW) characteristics and the influence of these variable values on the anaerobic digestion (AD) process, FW characteristics from 70 papers were compiled and analysed statistically. Results indicated that FW characteristics values are effectively very variable and that 24% of these variations may be explained by the geographical origin, the type of collection source and the season of the collection. Considering the whole range of values for physicochemical characteristics (especially volatile solids (VS), chemical oxygen demand (COD) and biomethane potential (BMP)), FW show good potential for AD treatment. However, the high carbohydrates contents (36.4%VS) and the low pH (5.1) might cause inhibitions by the rapid acidification of the digesters. As regards the variation of FW characteristics, FW categories were proposed. Moreover, the adequacy of FW characteristics with AD treatment was discussed. Four FW categories were identified with critical characteristics values for AD performance: (1) the high dry matter (DM) and total ammonia nitrogen (TAN) content of FW collected with green waste, (2) the high cellulose (CEL) content of FW from the organic fraction of municipal solid waste, (3) the low carbon-to-nitrogen (C/N) ratio of FW collected during summer, (4) the high value of TAN and Na of FW from Asia. For these cases, an aerobic pre-treatment or a corrective treatment seems to be advised to avoid instabilities along the digestion. Finally, the results of this review-paper provide a data basis of values for FW characteristics that could be used for AD process design and environmental assessment.

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1. Introduction

Worldwide, about 1.3 billion tons of food is lost or wasted per year, which represents one-third of production in the food supply chain for human consumption (Gustavsson et al., 2011). According to the Food and Agriculture Organisation of the United Nations (FAO) (Jan et al., 2013), the direct cost of wastage of agricultural food products (excluding fish and seafood), based on the year 2009 producer prices, is about USD 750 billion. The carbon footprint of food produced and not eaten is estimated to be 3.3 $GTCO_2Eq$, which could be ranked, if food wastage was a country, as the third top greenhouse gases emitter, after China and USA. The European Commission estimated that about 89 MT (170 $MTCO_2Eq$) of food were lost or wasted in Europe in 2006, equivalent to 179 kg/capita/year. Moreover, 42% were generated by households (Monier et al., 2010). Food loss refers to the decrease in edible food mass at the early stages of the food supply chain, such as production and postharvest handling, and occurs mostly in developing countries. Food waste (FW), also called kitchen waste, refer to the discard of foodstuff, mainly composed of unsold food, food preparation leftovers and uneaten food from households, restaurants and large producers as collective caterers and supermarkets (Tom and Hannah, 2009). FW are more typical of lifestyles observed in high-income countries (Fig. 1) (Jan et al., 2013).

Nowadays, more FW reach landfills than any other single material (21.3% of the municipal solid waste in 2011 in USA (EPA, 2011)). Once in landfills, FW follow a natural biodegradation, principally in anaerobic way, releasing environmental harmful elements as leachates and gases, concentrated chemical oxygen demand (COD), ammonia, volatile fatty acids (VFA), and methane (CH_4) (Ağdağ and Sponza, 2007). That is why landfills are the third largest anthropogenic source of CH_4 in the world, representing about 11% of estimated total emissions, after emissions linked to fossil fuels (production, distribution and combustion) (33%), and enteric fermentation of livestocks, representing about 27% of the emissions (Bousquet et al., 2006; EPA, 2012). Several countries are gradually introducing limitations for FW landfilling, setting national objectives of FW collection and valorisation. According to the European Council Directive (EC, 1999), the biodegradable

* Corresponding author.
E-mail address: anne.tremier@irstea.fr (A. Tremier).

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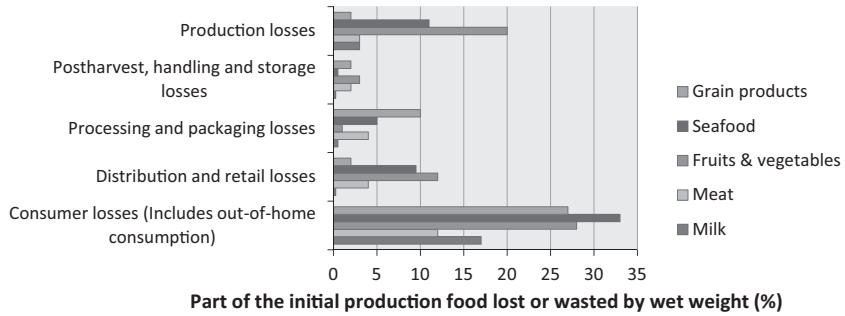


Fig. 1. Food waste in the food supply chain for USA, Canada, Australia and New Zealand (Gustavsson et al., 2011).

municipal waste going to landfills in 2016 must be reduced to 35% of the total amount (by weight) of the one produced in 1995. Additionally, each country has its own targets, as Belgium and Germany, which FW valorisation objectives are 75% by 2015 and 70% by 2020 respectively (Sidaine and Gass, 2013). In the same way, producers and holders of over 10 tons per year of FW in France will have to implement a source-separation management for a subsequent valorisation starting from 2016 (Ministere de l'écologie, 2011). More strictly, some countries like Scotland (from 2021 (Sidaine and Gass, 2013)) and South Korea (since 2005 (Cho et al., 2013)) are seeking to prohibit the direct landfilling of FW.

The high organic matter and the water content of FW, which mainly cause the harmful elements in landfills, can be valorised by biological treatments. Traditionally, the recovery of the biological value of FW is led by composting but incineration is also performed on FW, mixed with dry waste to raise their calorific value (EC, 2003, 2009; Zhang et al., 2007). Nowadays, global targets claim the need for alternative and energetically more efficient routes of valorisation. Among the biological valorisation, the anaerobic digestion (AD) of FW has demonstrated to be one of the most advantageous technologies to maximise the substrate and energy recovery (Castillo et al., 2006; Forster-Carneiro et al., 2008; Fricke et al., 2007; Hartmann and Ahring, 2006; Krishna and Kalamdhad, 2014). However, to respond to the current extensive need of FW valorisation, AD treatments have to fulfil essential conditions such as robustness, high adaptive capacity and good economic performances. The performances of AD processes of FW are largely influenced by the substrate characteristics and kinetics of degradation (Carlsson et al., 2012). But so far, no clear information is available, neither about the FW characteristics variations caused by the different sources, nor about the possible effects of these variations on AD.

The aim of this review-paper is thus to give an overview of the general mean values of FW characteristics, to highlight the origins of their major variations and to discuss the AD behaviour against these variations. In this objective, a statistical analysis was applied on FW characteristics collected from scientific papers studying various techniques of valorisation of FW throughout the world. The results will improve the knowledge on FW characteristics, provide some advices for a more efficient valorisation of FW making use of AD and supply more accurate data for environmental assessment, simulation and predesign of AD with FW as feedstock.

2. Method

2.1. Collection of literature data

In order to make an analytical description, FW characteristics from 102 different samples, proposed in 70 studies conducted throughout the world between 2001 and 2014, have been extracted from scientific databases such as Scopus® and Web of Science™. The bibliographical references of FW characteristics used are listed in the appendix.

2.2. Studied FW characteristics

All characteristics gathered were classified in physicochemical characteristics, elementary composition, biochemical characteristics, nutritional elements and heavy metals. Only characteristics with at least 9 available data were considered in the present study in order to keep a sufficient representativeness.

The definition of some characteristics may be different among publications. For this paper purpose, it was chosen to aggregate them in the same unit as shown in Table 1. For example, some researchers told about the organic matter (OM) rather than of the volatile solids (VS), but in the present study it was chosen to gather both type of data as VS. Similarly, total sugars and carbohydrates were assembled together as carbohydrates. Carbohydrates value does not include structural carbohydrates, as cellulose (CEL) and hemicellulose (HEM). Additionally we studied separately the total COD and the water-soluble COD (CODs). Some authors took the total nitrogen content (N) as for the total Kjeldahl nitrogen (TKN), assuming a negligible amount of nitrites (NO_2–N) and nitrates (NO_3–N) in FW (Adhikari et al., 2013). In the present study, N and TKN were taken separately because of nitrites and nitrates evidenced in some studies (Seo et al., 2004; Sundberg et al., 2011). Moreover, some authors made the difference between ammonium (NH_4^+) and ammonia (NH_3), but, depending on the pH, NH_4^+ and NH_3 can be transformed into one form or the other (Anthonisen et al., 1976). All data of NH_4^+ and NH_3 were thus aggregated as total ammonia nitrogen (TAN). The carbon-to-nitrogen ratio (C/N) was calculated in some studies with N or TKN indistinctly without a clear notification, thus it was not possible to make the distinction in this paper. Additionally, some minerals like Fe and Zn were either considered as heavy metals or nutritional elements. In the present study they were included in nutritional elements.

2.3. Information characterizing data origins

In order to perform a statistical analysis, FW characteristics data were sorted considering three categories of possible variation factors: continents of the analysed FW, collection sources and season of collection.

In the category "continent", three main subcategories were chosen: Asia, North America (NA) and European Union (EU), with a proportion of data of 42%, 15% and 43% respectively. The availability of data from Africa, Oceania and Central and South America was very limited. Even gathering the data available for these latter continents, the number of data was still insufficient to make them statistically representative. Thus they were not included in this study.

Table 1
Resume of food waste characteristics from 102 samples (standard deviations in parenthesis).

Parameters	n	$\bar{x}$ (σ)	CV (%)	Std. Skew.	Std. Kurt.	CI
Physicochemical characteristics						
pH	65	5.1 (0.7)	13.9	1.3	−0.7	0.2
DM (%WW)	94	22.8 (10.0)	43.7	2.3	1.4	2.0
VS (%DM)	73	88.2 (8.2)	9.3	−6.0	5.5	1.9
COD (g/kg WW)	15	198 (89)	45.0	0.4	−1.2	49.4
CODs (g/kg WW)	14	99 (157)	158.7	5.0	8.9	163.9
BMP ($NLCH_4$/kg VS)	10	460.0 (87.6)	19.0	−0.4	−0.9	62.6
Elementary composition						
C (%DM)	68	45.5 (7.5)	16.6	−1.5	5.6	1.8
H (%DM)	28	7.0 (3.3)	46.5	7.6	16.4	1.3
O (%DM)	21	33.2 (6.7)	20.1	−2.4	1.5	3.0
N (%DM)	79	2.8 (1.1)	40.3	4.7	4.7	0.3
S (%DM)	23	0.4 (0.2)	56.9	1.7	−0.2	0.1
C/N	68	18.5 (5.9)	31.8	3.4	1.8	1.4
Biochemical characteristics						
Carbohydrates (% VS)	24	36 (21)	57.2	−0.2	−0.8	8.8
HEM (%VS)	9	9 (5)	52.8	0.9	0.3	3.8
CEL (%VS)	10	9 (8)	94.5	1.3	−0.1	5.8
LIG (%VS)	11	7 (6)	92.8	1.5	0.2	4.1
Proteins (%VS)	26	21 (13)	62.2	5.9	13.2	5.3
Fats (%VS)	28	15 (8)	52.0	3.53	4.54	3.1
Nutritional elements						
TKN (g/kg DM)	30	16 (12)	75.8	0.6	−1.5	4.5
TAN (mg/L)	35	731 (958)	131.0	6.1	8.4	317.5
NO_3–N (ppm DM)	9	105 (50)	47.4	1.7	2.1	348.1
Ca (%DM)	12	1.6 (1.3)	85.5	0.2	−1.3	0.9
K (%DM)	24	1.2 (0.7)	57.7	1.6	0.7	0.3
Mg (%DM)	11	0.2 (0.2)	116.4	2.8	2.8	0.2
Na (%DM)	12	2.2 (2.9)	132.3	2.5	1.2	1.8
P (%DM)	33	0.5 (0.3)	62.9	2.6	1.6	0.1
Fe (ppm DM)	16	482 (815)	168.9	5.0	8.6	431.9
Mn (ppm DM)	15	62.5 (48.4)	77.5	1.1	0.4	26.7
Mo (ppm DM)	10	1.2 (1.1)	90.2	1.1	−0.2	0.8
Zn (ppm DM)	18	87 (110)	127.4	3.4	3.1	54.7
Heavy metals						
Cd (ppm DM)	12	0.3 (0.4)	116.5	3.6	5.3	0.3
Cr (ppm DM)	13	28 (71)	258.1	5.1	9.1	42.8
Cu (ppm DM)	16	23 (37)	157.9	3.6	3.2	19.7
Hg (ppm DM)	9	0.3 (0.4)	157.0	1.9	0.4	0.3
Ni (ppm DM)	14	10 (25)	255.9	5.6	10.3	14.5
Pb (ppm DM)	12	18 (46)	252.4	4.6	7.6	29.0

For the category "collection source", five subcategories were chosen: restaurant food waste (RFW), household food waste (HFW), FW mixed with green waste (FWGW), FW of large producers (FWLP) and organic fraction of municipal solid waste (OFMSW). RFW represents the most important part of the available data with 41% of the total, as shown in Fig. 2.

Most of the authors described that FW from restaurants and households were source sorted. The subcategory FWGW concerns FW of different source (RFW, HFW, FWLP and OFMSW) that were mixed with green waste (GW). The FWLP are mainly constituted by unsold food products of supermarkets and hypermarkets. Few authors described the procedure for homogenization and packaging deconditioning of FWLP, so these procedures were not listed in the present study. Lastly, OFMSW represent all organic waste sorted (mechanically or by other methods) after mixed collection of municipal solid waste (MSW).

Finally for the category "season", the available data were very limited, as only 15 studies specified the season of FW collection. Nevertheless, 2 subcategories were identified: summer and winter, with a proportion of data of 47% and 53% respectively.

2.4. Statistical analysis

2.4.1. Representativeness of the mean value

A statistical study was performed using Statgraphics Centurion XVI® software. All data were characterized with the mean ($\bar{x}$), standard deviation (σ), coefficient of variation (CV) and a confidence interval (CI), as shown in Table 1. The normality of the distribution was tested by calculating the standardized skewness and the standardized kurtosis (Baillargeon, 2008).

For Martin and Gendron (2003), a CV between 0% and 16% shows a low variation on the sample. With a CV between 16% and 33%, the sample has an important variation and the mean may contain many errors. Finally with a CV over 33%, the variation in the sample is too high to consider the mean as representative for the whole population. In our case, to determine whether a FW characteristic mean is representative from the data set, this later must have a low CV (close or below 16%) and follow a normal distribution.

2.4.2. Variance analysis

For the case where the mean characteristic value could not be considered as representative, the Kruskal–Wallis test was used to compare the medians of FW characteristics in order to test the variation caused by the three categories ("continent", "collection source" and "season"). However, interactions between the categories were not studied. With a Kruskal–Wallis probabilities (KWP) lower than 0.05, a link between the variation of the characteristics and the selected categories is probable with certitude of 95%. In such cases, the Bonferroni correction was used to specify the subcategories that may generate these variations. For each characteristic, the subcategories are defined as belonging to a statistical group (1, 2 or 3). When several subcategories belong to the same group, significant statistical difference cannot be observed

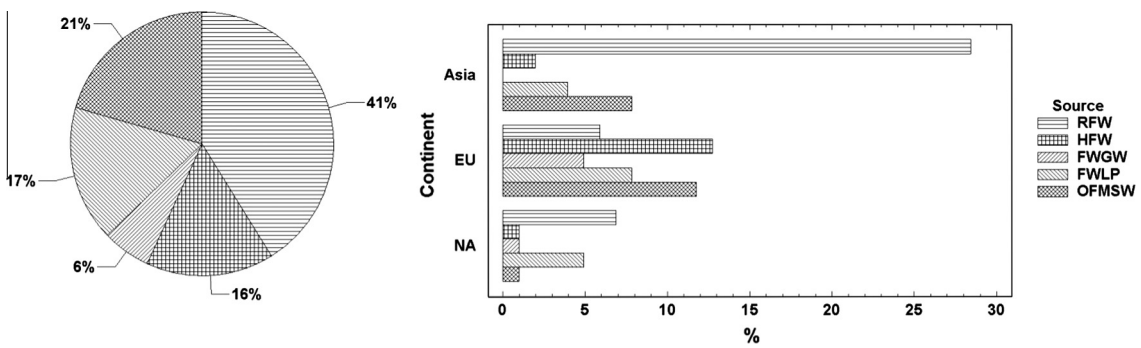


Fig. 2. Distribution of food waste source in total and by continents (% of data).

amongst themselves. To implement the Bonferroni correction, the data set was segregated into each individual subcategory. Only, the subcategories with at least 3 data were taken into account.

3. Results of the statistical review of FW characteristics

3.1. Mean characteristics and representativeness

Average data and CV for FW characteristics are presented in Table 1. In this section, a comparison between the FW and other substrates characteristics is proposed and the representativeness of FW characteristics values is discussed.

With a mean pH of 5.1, FW is more acid than other organic waste as green waste with a pH around 7.3 (Zhang and Sun, 2014), cattle manure with 8.7 (Bah et al., 2014) and sewage sludge with 7.8 (Martín et al., 2015). pH presents a CV lower than 16% and follows a normal distribution, inducing that its mean real value ranges (considering the CI) between 4.9 and 5.3 with a real σ between 0.6 and 0.8. These mean value and range of variation of pH can be universally used for process design or environmental assessment.

The biomethane potential (BMP) of 460.0 $NLCH_4$/kg VS (±87.6) is higher than that of most of organic waste: BMP from cattle manure and sludge sewage are 250 and 270 $NLCH_4$/kg VS respectively (Lesteur et al., 2010). BMP values for FW follow a normal distribution with a CV of 19% that is close to being considered as low. Using the CI, the BMP real mean ranges between 397.3 and 522.6 $mLCH_4$/g VS and the real σ between 60.2 and 159.3. These values can be reasonably used, whatever the type of FW, for first simulation of biodegradable potential for example.

Neither of the other mean values for physicochemical, biochemical or elementary characteristics can be considered as representative of the whole FW types. Indeed they present either CV beyond 33% or do not follow a normal distribution. As an example, VS presents a low CV (9.3%), but it is not considered as representative because it does not follow a normal distribution. Because of these variations, the comparison of these FW characteristics mean values with other waste makes no sense.

In general, FW prove to have varying proportions of nutriments and micronutrients and low presence of heavy metals. Nevertheless, once again these characteristics present very high CV (some of them exceed 100% and even 200%) and do not follow normal distribution.

The causes of variations and non-normality were searched and are presented in the next section in order to provide a better understanding of FW characteristics variability and to identify more representative means depending on chosen categories.

3.2. Impact of FW categories on FW quality variation

The Kruskal–Wallis test was realized for all FW characteristics, except pH and BMP. Variations observed for nine characteristics demonstrated to be significantly influenced by the three studied categories (Table 2): geographical origin, collection source, and season.

Three characteristics showed to be impacted by the geographical origin of the FW: dry matter (DM), TAN and Na content. In comparison to DM general mean (22.8%), FW from Asia and NA present lower values (18.5% and 18.1% respectively) than DM of FW from EU (23.7%). TAN value of FW from Asia shows to be lower than the TAN means of FW from EU and NA (249 against 989 and 1434 mg/L respectively). Meanwhile, Na values for FW from Asia shows to be the highest (5.3 against 0.5 and 0.9%DM, respectively for EU and NA).

The category "collection source" had the most frequent impact on FW characteristics with six characteristics impacted: DM, VS, CEL, carbon (C) content, oxygen (O) content and TAN content. For these characteristics, no significant statistical difference was evidenced between the FW collected from restaurants, households and large producers (respectively RFW, HFW and FWLP) as shown in Table 2. On the contrary FW issued from MSW (OFMSW) or collected with mixed green waste (FWGW) demonstrated significant differences with the other collection source sub-categories. Concerning FWGW, DM (39.4%WW) and TAN (2627 mg/L) are higher and the C content (35.8%DM) is lower than those of the general mean value. The presence of GW in FWGW seems to be a very important factor of variation for FW characteristics. Regarding OFMSW, a high content of CEL (25.2%VS) and a low VS (82.1% DM) and O content (27.5%DM) are stated. This difference can highlight the difficulty to separate the napkins, papers and cardboards, and the higher presence of impurities in OFMSW, which may explain these variations.

The "season" category explained the variations of only three characteristics: C, N and C/N. A lower C content (46.4%DM) and a high N content (3.8%DM) are evidenced in FW collected in summer. Consequently a lower C/N ratio (13.6) is evidenced too. It must be recalled here that data for "season category" covered different types of collection sources. Thus in the same way as FWGW, these differences on C and N contents are probably caused by a higher supply of GW (or assimilated waste as vegetables peels) in FW collected during summer. This assumption is supported by the GW characteristics that havea higher DM, N and TAN content and a lower C content than FW only (Chen et al., 2014). This may explain the differences observed in FWGW and in FW collected in summer.

In Table 3 are presented the specific mean values of each characteristic impacted by the selected categories. With these values it is possible to select, with more accuracy, the value of each specific FW characteristic according to its geographical localisation, collection source and season of collection. It is especially interesting when data are needed to design a process or to supply an environmental assessment analysis. However, the variation factors studied in this paper (FW categories) explain the variations for only 9 of the 37 FW characteristics (24% of the studied characteristics). Thus other studies would be necessary to clarify the origins of the other characteristics variations. Moreover, interactions between categories were not investigated because of a lack of relevant data.

4. Discussion: Pros and cons of FW characteristics for valorisation through AD

4.1. Recall of AD fundamentals and influencing factors

AD is a biological treatment which can lead to energy production, pollution control and fertilising substrate recovery from organic waste. Microorganisms that produce CH_4 utilise a limited number of substrates, mainly acetate, dihydrogen (H_2) and carbon dioxide (CO_2). Consequently, a decomposition of raw substrate is necessary to provide these elements to the microorganisms. As well known, AD follows four key steps: hydrolysis, acidogenesis, acetogenesis and methanogenesis. These steps tend to work more or less simultaneously depending on the availability of degraded molecules, environmental conditions and biomass development. Several studies already reviewed the fundamentals of AD, the environmental conditions and the inhibitors of the process (Deublein and Steinhauser, 2011). Thus this section only aims at summarizing the major phases of AD of OM and their main influencing factors in order to build links with the anaerobic treatability of FW.

In the hydrolysis step, the complex composite particles and particulate carbohydrates, proteins and lipids are broken down in

Table 2
Kruskal–Wallis probability and Bonferroni groups for significant variations sources.

Parameters	Continent				Collection source						Seasons		
	KWP	Asia	EU	NA	KWP	RFW	HFW	FWGW	FWLP	OFMSW	KWP	Summer	Winter
Physicochemical characteristics													
DM (%WW)	0.00	1	2	1	0.00	1	1	2	1	1	0.06	–	–
VS (%DM)	0.19	–	–	–	0.01	1	1/2	1/2	1/2	2	nd	nd	nd
Elementary composition													
C (%DM)	0.72	–	–	–	0.01	1	1/2	3	1/2	2/3	0.02	1	2
O (%DM)	0.20	–	–	–	0.01	1	1/2	1/2	1/2	2	nd	nd	nd
N (%DM)	0.24	–	–	–	0.16	–	–	–	–	–	0.04	1	2
C/N	0.42	–	–	–	0.12	–	–	–	–	–	0.01	1	2
Biochemical characteristics													
CEL (%VS)	0.10	–	–	–	0.02	1/2	1	nd	1/2	2	nd	nd	nd
Nutritional elements													
TAN (mg/L)	0.00	1	2	2	0.01	1	1	2	1	1	0.31	–	–
Na (%DM)	0.01	1	2	2	0.07	–	–	–	–	–	nd	nd	nd

KWP: Kruskal–Wallis Probability, nd: not determined, –: without significant variations. Only characteristics which showed a KWP lower than 0.05, in some subcategories, are presented in this table. Subcategories that belong to the same group for FW characteristics proved to have no significant statistical differences between themselves.

Table 3
Resume of FW characteristics means including subcategories variations (standard deviations in parenthesis).

Parameters	$\bar{x}$ (σ)	Continent means			Collection source means					Seasons means	
		Asia	EU	NA	RFW	HFW	FWGW	FWLP	OFMSW	Summer	Winter
Physicochemical characteristics											
pH	5.1 (0.7)	–	–	–	–	–	–	–	–	–	–
DM (%WW)	22.8 (10.0)	18.5 (7.1)	28.4 (9.8)	18.1 (9.7)	19.1 (6.8)	23.4 (6.7)	39.4 (8.4)	23.1 (7.7)	24.0 (14.1)	–	–
VS (%DM)	88 (8)	–	–	–	91.6 (4.1)	–	–	–	82.1 (12.3)	nd	nd
COD (g/kg WW)	198 (89)	–	–	nd	–	nd	nd	–	–	nd	nd
CODs (g/kg WW)	99 (157)	–	–	nd	–	–	nd	–	–	nd	nd
BMP ($mLCH_4$/g VS)	460 (88)	–	–	nd	–	–	nd	–	nd	nd	nd
Elementary composition											
C (%DM)	45.5 (7.5)	–	–	–	48.6 (7.1)	48.0 (2.7)	35.8 (12.1)	46.5 (2.4)	41.0 (8.1)	46.4 (1.4)	48.4 (0.8)
H (%DM)	7.0 (3.3)	–	–	–	–	–	nd	–	–	nd	nd
O (%DM)	33.2 (6.7)	–	–	–	39.1 (1.0)	–	–	–	27.5 (8.85)	nd	nd
N (%DM)	2.8 (1.1)	–	–	–	–	–	nd	–	–	3.8 (1.7)	2.4 (0.5)
S (%DM)	0.4 (0.2)	–	–	–	–	–	nd	–	–	nd	nd
C/N	18.5 (5.9)	–	–	–	–	–	–	–	–	13.6 (4.7)	20.5 (4.5)
Biochemical characteristics											
Carbohydrates (%VS)	36 (21)	–	–	–	–	–	nd	nd	–	nd	nd
HEM (%VS)	9 (5)	nd	–	nd	–	–	nd	nd	nd	nd	nd
CEL (%VS)	9 (88)	–	–	nd	–	3.1 (1.76)	nd	–	25.2	nd	nd
Lignin (%VS)	7 (6)	–	–	nd	–	–	nd	nd	nd	nd	nd
Proteins (%VS)	21 (13)	–	–	nd	–	–	nd	–	–	nd	nd
Fats (%VS)	15 (8)	–	–	nd	–	–	nd	–	–	nd	nd
Nutritional elements											
TKN (g/kg DM)	16 (12)	–	–	–	–	–	–	–	–	–	–
TAN (mg/L)	731 (958)	249 (286)	989 (1190)	1434 (402)	483 (569)	310 (259)	2627 (1690)	772 (449)	193 (210)	–	–
NO_3–N (ppm DM)	105 (50)	nd	–	nd	nd	nd	–	–	nd	nd	nd
Ca (%DM)	1.6 (1.3)	–	–	–	–	–	nd	–	nd	nd	nd
K (%DM)	1.2 (0.7)	–	–	–	–	–	nd	–	–	nd	nd
Mg (%DM)	0.2 (0.2)	Nd	–	–	–	–	nd	–	nd	nd	nd
Na (%DM)	2.2 (2.9)	5.3 (3.5)	0.5 (0.2)	0.9 (0.4)	–	–	nd	nd	nd	nd	nd
P (%DM)	0.5 (0.3)	–	–	–	–	–	–	–	–	nd	nd
Fe (ppm DM)	482 (815)	nd	–	–	–	–	nd	–	3300	nd	nd
Mn (ppm DM)	62 (48)	nd	–	–	–	–	nd	–	nd	nd	nd
Mo (ppm DM)	1.2 (1.1)	nd	–	–	–	–	nd	nd	nd	nd	nd
Zn (ppm DM)	87 (110)	nd	–	nd	–	–	nd	–	250.7 (165)	nd	nd
Heavy metals											
Cd (ppm DM)	0.3 (0.4)	nd	–	nd	nd	–	nd	nd	–	nd	nd
Cr (ppm DM)	28 (71)	nd	–	–	nd	–	nd	nd	–	nd	nd
Cu (ppm DM)	23 (37)	nd	–	–	–	–	nd	–	–	nd	nd
Hg (ppm DM)	0.3 (0.4)	nd	–	nd	nd	–	nd	nd	nd	nd	nd
Ni (ppm DM)	10 (25)	nd	–	–	nd	–	nd	–	–	nd	nd
Pb (ppm DM)	18 (46)	nd	–	–	nd	–	nd	nd	–	nd	nd

Where: nd not determined, – without significant variations.

water soluble monomers as glucose, amino acids, and long chain fatty acids (LCFA) (Batstone et al., 2002). This step follows variable kinetics, depending on the complexity of OM. This degradation is achieved mainly by bacteria secreting extracellular enzymes. The

facultative anaerobic bacteria consume the remaining available oxygen after establishing anaerobic conditions, decreasing the reduction potential. It helps reaching the optimal conditions for the strictly anaerobic bacteria. In the acidogenesis step the monomers produced in the hydrolysis step are degraded in VFA, lactate, ethanol, H_2 and CO_2. This transformation is performed by hydrolytic fermentative bacteria (facultative anaerobic organism), which are also present in the hydrolysis step. However, this acidogenesis step has a faster kinetics than hydrolysis. In the acetogenesis step, the metabolites from hydrolysis and acidogenesis steps are transformed mainly in acetate (CH_3COOH or $C_2H_4O_2$), H_2 and CO_2. Two groups of acetogens may be identified: the obligate hydrogen producing acetogens, which degrade VFA, LCFA, alcohols and aromatics compounds in acetate and the homoacetogens, which produce acetate from H_2 and CO_2 (Anderson et al., 2003). Finally, the methanogenesis is the step responsible of CH_4 production which remains distinct from the other steps because the methanogens microorganisms are not bacteria but *Archaea*. This step can only work in strict anaerobic conditions.

Because of these complex succession of biological steps, AD of OM is a process with a more delicate balance than aerobic digestions, making the instability of digesters a common problem. In order to avoid instabilities, the knowledge and control of environmental and physicochemical factors influencing AD process are essential. In addition, in order to keep steady state, the digester must not be subjected to sudden changes of operating conditions (surcharges, variations of alimentation composition, temperature and pH ...). Influencing factors that are linked to the characteristics of the substrate to be introduced in the digester are listed below.

pH is one of the most essential parameter for AD, because of a sensitivity of microorganisms to pH variations. However, each step of the process shows a different pH sensitivity. A pH between 7.0 and 8.0 is suitable for protein degradation and between 6.0 and 9.0 for carbohydrate degradation (Walker et al., 2010). For fermentative bacteria, a comprehensive pH range from 4.0 to 8.5 is suitable. Most methanogens work optimally in a pH range from 6.5 to 7.2 (Zhang et al., 2014), with a methanogenesis step failure below pH of 5.5 and a limitation above a pH of 8.0. pH is highly linked with TAN and VFA concentration, making these elements the major parameters of the pH balance management.

TAN is produced by the biological degradation of amino acids, proteins and urea. TAN concentration is beneficial to methanogens at low concentrations (<200 mg/L or 12 mM (Liu and Sung, 2002; Walker et al., 2010)) but in high concentration is inhibitory. It was found that concentrations above 2500 mg/L led to 50% inhibition of methanogenic activity in several substrates (Benabdallah El Hadj et al., 2009; Chen et al., 2008; Zhang et al., 2012). According to Poggi-Varaldo et al. (1997), different concentrations of TAN in digester can be inhibitory in mesophilic conditions depending on substrate nature: cow manure with TAN > 3000 mg/L, agro-industrial wastes with TAN > 5000 mg/L, sewage sludge with TAN > 5000 mg/L, OFMSW with TAN > 2500 mg/L and FW with TAN > 3500 mg/L. However, in thermophilic conditions, the process can be inhibited by TAN concentrations in digester of above 1700 mg/L (Chen et al., 2008) whatever the substrate. Process instability caused by TAN excess often results in VFA accumulation (Chen et al., 2008), decreasing the pH.

As mentioned previously, VFA are produced in the acidogenesis step from monomers (monosaccharides, amino acids, long chain fatty acids). Monomers high availability produces a rapid VFA production. If acetogen microorganisms are not sufficiently developed to consume the available VFA, for example because of inhibition by TAN concentrations, then VFA may accumulate in the digester. High stress in AD was found with VFA concentration by 20 g/L (Zhang et al., 2012) and a complete inhibition of the acetogenesis step was observed by a concentration of 40–50 g COD/L (equivalent to 25–31 g/L of VFA) (Veeken and Hamelers, 2000). Anyhow, a right balance between carbohydrates and proteins contents and the correct pH control are essential to avoid inhibition from TAN and VFA concentrations.

To ensure the microorganisms activity, a good nutritional balance is necessary. The C and N are the principal elements consumed by microorganisms, which need between 15 and 30 times more C than N. In general, AD works fine with a C/N ratio between 20 and 30 (Teglia et al., 2011), though, it was recently observed that the digestion proceeds well at low C/N between 15 and 20 (Zhang et al., 2014). In addition, many micronutrients as Ca, K, Mg, Na, P and Fe are indispensable for microorganisms. A compilation of the function of each micronutrient in AD was proposed by Walker et al. (2010). The optimal values of micronutrients are difficult to calculate because the bioavailability depends on multiple factors which are specific to each substrate and configuration.

Concerning heavy metals (Cu^{2+}, Pb^{2+}, Cd^{2+}, Ni^{2+}, Zn^{2+}, Cr^{6+}), inhibitions have been observed in AD. But, as for the micronutrients, the limits are very variable and depend of reactor conditions and configuration. In addition, the definition of heavy metals differs among the studies, making difficult to compare the inhibition values. According to Lin (1992), the relative toxicity of heavy metals to degradation of acetic acid in mesophilic AD of sewage sludge was found to be Cd > Cu > Cr = Zn > Pb > Ni.

Finally, the initial DM content of the substrate may determine the additional water needs in AD and the most suitable AD technique. Dry and wet AD conditions are available to treat solid substrates. Dry AD facilitates the dry substrate manipulation, with batch and semi-continuous process but higher structure cost may be present. In this case, the desirable initial DM is between 20% and 40%, which allows performing the AD without important water addition. Wet AD facilitates the contact between the microorganisms and the substrate with continuous and completely mixed process. But in this case there are higher water requirements to adjust the substrate initial DM content. Indeed, the desirable DM content within a wet digester is under 20%.

4.2. Comparison between FW quality and optimal conditions for AD treatment

Considering the whole range of values for physicochemical characteristics (especially VS, COD and BMP) shown in Table 3, FW show a good potential for AD treatment. However, within the scientific literature dealing with AD of food waste, several authors highlighted the occurrence of instabilities in the treatment that cause the falls of pH, a lower gas yield and the increase of CO_2 content in biogas (Mata-Alvarez et al., 1992). Traditionally, remedial measures are applied as alkali addition, feed interruption and mixing with a nitrogen-rich supplement (Jiang et al., 2012). The results of the present study allow proposing appropriate AD configurations to avoid instabilities linked to the FW characteristics values as a function of different categories, and to improve FW adequacy with AD treatment.

The statistical analysis (Section 3.1) stated that the initial pH of FW is 5.1 (±0.7) whatever the FW categories. Even if this range of pH is favourable for fermentative bacteria that could easily develop in the first days of the AD process, a higher pH is necessary in the digester in order to favour the development of methanogen microorganisms. Moreover, this low pH combined with the potentially high carbohydrates and proteins contents of FW (respectively 36.4%VS (±20.8) and 21.0%VS (±13.0)), may explain the rapid acidification of reactors, caused mainly by the high production of VFA and the high concentration of NH_4^+ in the digesters (Deublein and Steinhauser, 2011; Mata-Álvarez, 2003; Scano et al., 2014). Consequently, inhibitions of acetogenesis and methanogenesis steps could be evidenced. On the other hand, the structural

carbohydrates contents (CEL with and HEM) may be higher than 8.7% and 9.4% of VS respectively and may induce an incomplete degradation of the OM (only 50% of CEL is decomposed during AD at 37 °C for 30 days) (Haug, 1993). Concerning micro-nutriments, Ca (1.6%DM (±1.3)), K (1.2%DM (±0.7)), Mg (0.2%DM (±0.2)), Na (2.2%DM (±2.9)), P (0.5%DM (±0.3)) and Fe (482.5 ppm (±815) are present in FW and, according to Walker et al. (2010), their concentrations are close to those recommended to AD. Only the nitrogen/phosphorus ratio might sometimes be lower than the value recommended by Walker to improve the AD process (N/P = 7).

Looking more accurately at the characteristic values found per FW categories, some additional attention points may be highlighted according to the FW geographical origin on one hand, and the FW producer and collection sources on the other hand.

Concerning the geographical origin, the statistical study showed that:

- TAN concentration for Asia (249 mg/L) is considered close to the optimal concentration (Liu and Sung, 2002) and TAN concentration from NA (1434 mg/L) and EU (989 mg/L) are higher but not considered as inhibitory as discussed previously.
- The high Na content in Asian FW (5.3%DM) can be considered as moderately inhibitory of AD (Moletta, 2011), thus a special attention is advised in this continent.

Concerning the producer and collection sources, the waste characteristics highlighted that:

- The mean value of DM in FWGW (39.4%, Table 3) is higher than that of the other subcategories. The treatment of such a FW in wet AD may be more difficult because of higher needs for additional water in the digester and because of mixing difficulties. Thus for such a collection type, a dry process might be preferable.
- TAN concentration of FWGW (2627 mg/L) is inhibitory and might cause a 50% reduction in the methane yield, even at low organic loading rate (OLR) (Benabdallah El Hadj et al., 2009; Zhang et al., 2012). Moreover, a process instability may be present due to VFA accumulation.
- Regarding structural carbohydrates, the high content of CEL in OFMSW (25.2%VS), compared to the global mean value (8.7%VS), may cause difficulties to calculate the OLR, because of the accumulation of recalcitrant OM. According to Diaz et al. (2011), the lack of oxygen supply in the early hydrolysis step, affects the AD of CEL, resulting in a production of biogas during the first days but an incomplete transformation of CEL.
- Finally, the minimum recommended C/N ratio (15) for AD, is not reached by FW in summer (13.6). As a consequence, failures may occur by excess of N, ending in TAN and VFA accumulation, or deficiencies of C in AD.

Considering the above analysis highlighting links between specific characteristics of FW categories and possible instabilities of an AD treatment, specific AD configurations or management attention points may be advised to limit these instabilities:

- In the case of FW with low DM, especially FW from NA with 18.1%, a wet AD seems to be beneficial to take advantage of the initial water content. On the contrary, in case of high DM as in FWGW, a water addition, a mixing with substrates with lower DM or a separation of GW of FW is advised to develop a correct AD process.
- Because of the FW low pH, the feeding of the digester must be carefully managed, in terms of OLR, in order to avoid decrease of pH in the digester.
- In the same way, a specific attention must be paid to the control of the TAN concentration, by dilution with water or by air stripping before FW introduction in the digester (Chen et al., 2008; Krishna and Kalamdhad, 2014).
- In all cases, GW content in FW seems to generate several difficulties in AD process: DM and TAN content. Thus a separation of the fraction of GW may be useful to avoid instabilities or a co-digestion with substrates with a low TAN content has to be advised. Such solutions will improve the AD of FWGW and of summer FW that can be affected by the low C/N of the substrate. Moreover for FWGW, composting treatment might to be preferred to AD to take advantage of high DM content and to avoid problems linked to long degradation time of green waste in AD.
- Finally, to avoid VFA accumulation, an enhancement of the acetogens and methanogens microorganisms may be promoted by inoculum addition to consume more rapidly the VFA produced. The carbohydrates content, that generates VFA, may also be partially eliminated by an aerobic pre-treatment. Such aerobic pre-treatment is also recommended to increase the exoenzymes production, improving CEL and other polysaccharides degradation (Bon and Ferrara, 2007). Charles et al. (2009) showed that higher production of exoenzymes enhance the hydrolysis step of AD of FW. Variations in the populations of microorganisms were stated depending of the oxygen availability in the hydrolysis step. As an example oxygen favours the development of CEL degrading bacteria (*bacteroidetes*) (Lim et al., 2014).

4.3. Lacks in terms of characterization of FW for a better understanding of AD stability

In this study high variations in FW characteristics were demonstrated. However, factors of variation that were studied (FW categories) explain the variations of only 9 FW characteristics. Therefore, other factors should be studied to identify other causes of variations of the FW characteristics. As an example, the typology of the FW compounds might be interesting since FW are composed by variable contents of fruits, vegetables, cereals, bakery, meat, dairy, among others (Heaven et al., 2013), as shown in Table 4. Furthermore, details about the different FW sorting or collection techniques or about the initial degradation degree caused by the FW storage before AD may be interesting to understand the variations of FW characteristics.

To complete the understanding of the effect of FW characteristics on AD, a thorough knowledge of the links between FW biochemical characteristics and biodegradability might be important. As an example, it was observed in the literature that the potentially high carbohydrates content in FW may cause instabilities (Scano et al., 2014), but a clear differentiation in the composition of these carbohydrates would be important to know their real impact in AD. Studies of composting degradation kinetics showed that water soluble organic particles with a size lower than

Table 4
Comparison of FW composition in UK, Finland, Portugal, Italy and the 2008 UK WRAP analysis (Heaven et al., 2013).

% Wet weight	UK	Finland	Portugal	Italy	UK (WRAP)
Fruit and vegetable waste	60.9	44.5	59.2	69.0	46.6
Pasta/rice/flour/cereals	1.5	0.4	0.2	12.4	2.5
Bread and bakery	9.0	3.8	3.1	2.8	13.4
Meat and fish	6.7	4.3	7.3	6.2	8.4
Dairy	1.7	2.0	0.7	1.4	3.5
Drinks	7.1	27.5	0.2	0.0	8.0
Confectionery, snacks, etc.	0.7	3.2	0.3	0.0	1.7
Mixed meals	12.3	6.3	29.0	1.4	12.9
Other food	0.2	8.0	0.0	6.9	3.0

1.5 kDa are more easily biodegradable than larger water soluble particles (Tremier et al., 2014). Such a fractionation of the OM might be useful to understand the hydrolysis step in AD and in particular in the case of the use of an aerobic pre-treatment.

5. Conclusion

Characteristics of FW generated throughout the world were compiled. The adequacy of these characteristics with valorisation was studied. AD is known to be one of the best technologies for energy and substrate valorisation of FW. However, the FW characteristics proved to be highly variable as stated in this study and some of them may generate instability in AD process. This study showed that only two FW characteristics have universal values with CV lower than or close to 16%: pH and BMP. The variations of 24% of the studied characteristics were explained by the geographical origin, the collection source and the collection season of FW. In particular, this study showed significant differences for: (1) characteristics of Asian FW against EU and NA; (2) characteristics of OFMSW and FWGW against RFW, HFW and LPFW and; (3) characteristics of FW collected in summer against FW collected in winter. FW characteristics values may affect their treatment through an AD process: the low pH, the high carbohydrates content, the high value of TAN and Na of FW especially from Asia; the high DM, and TAN from FWGW; the high CEL content of FW from OFMSW and the low C/N ratio of FW collected during summer. A corrective treatment during AD or an aerobic pre-treatment of FW seems to be advised to avoid instability in digesters caused by these characteristics. For the characteristics which variations could not be explained, other causes of variations have to be searched in the future: for example, the typology of FW, the sourced collected methods or the socio-economic situations of FW producers. In addition, other characteristics like the biodegradability kinetics, the biomass content, the enzymatic variations and the soluble OM fractionation may help to understand clearly the AD of FW and to propose perspectives of optimization of the process.

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Appendix

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